

ELEC 3115

Lecture 6: Week 3, Thursday

228 responses submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)

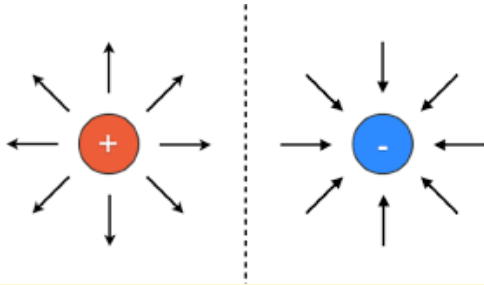
image method image charge Poisson equation equation and poisson
numerical method laplaces and poisson electrostatic problems
Gauss law **Method of image** Electric field
Applications for Laplace boundary problems
Poisson and laplace Poissons equations use Laplace equations different methods
equations and method Poissons and Laplace
analytical methods

228 responses submitted

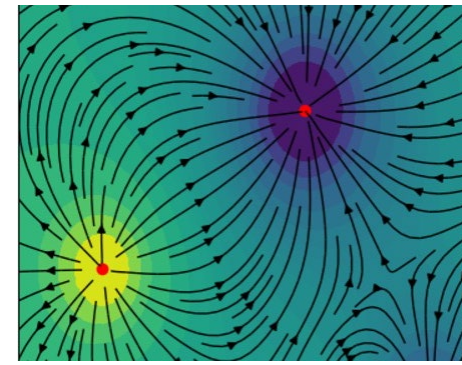
What is still unclear or confusing to you right now? (Responses should reflect your own understanding.)

solution method Image of two conductors image location
intuition behind the method Image charges image method Laplace Equation
methods under exam **method of image** Poissons equations
Mirror method variables or method
cylindrical conductor Boundary Conditions image placement
image work correct method use of image use the method maths and method

Review: Method of images



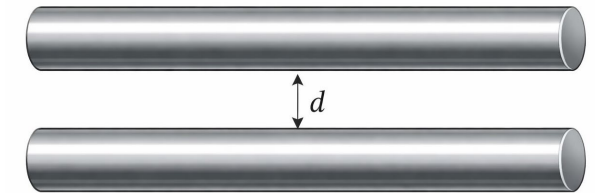
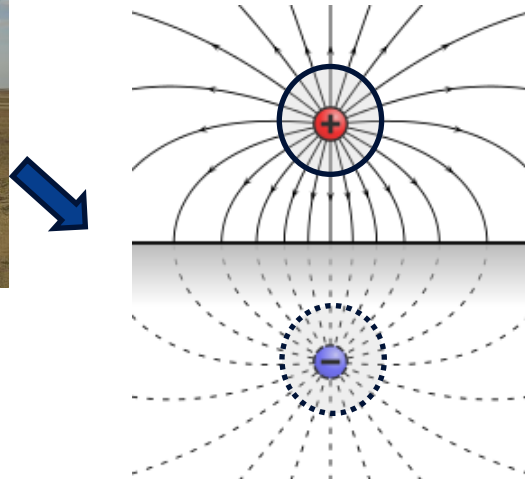
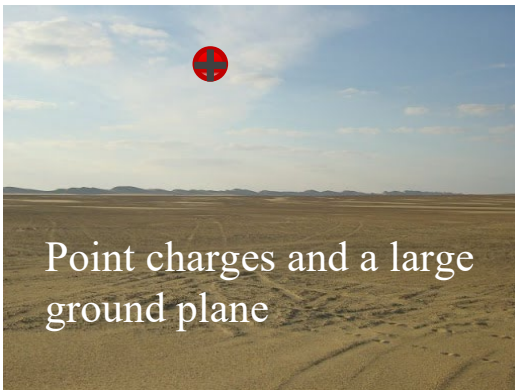
Gaussian surfaces enclosing the charged bodies can be found when they are far apart and do not affect electric field of each other.



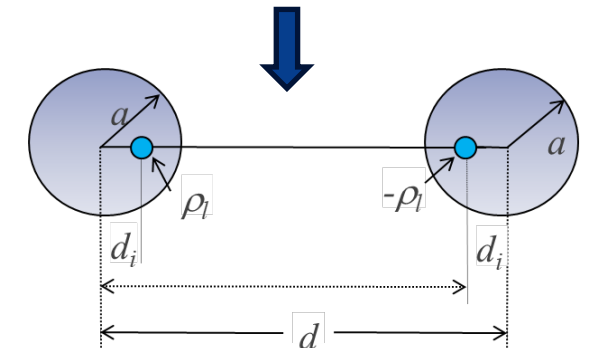
When Gauss's law is inapplicable, a problem may require solving Poisson's/Laplace's eqns (PDEs).

When the geometry is simple, an alternative is the method of images (representing a conductor as a mirror), which is possible because of the uniqueness theorem for PDEs (Partial Differential Equations).

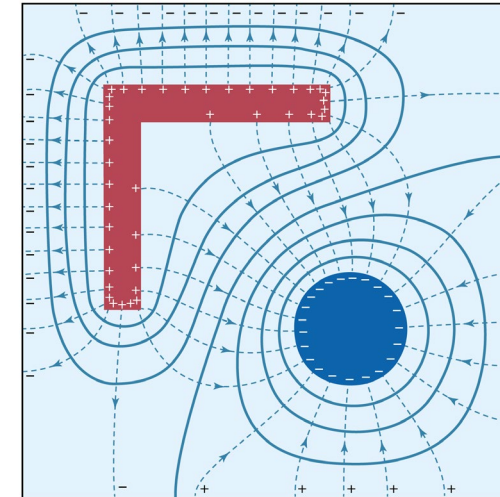
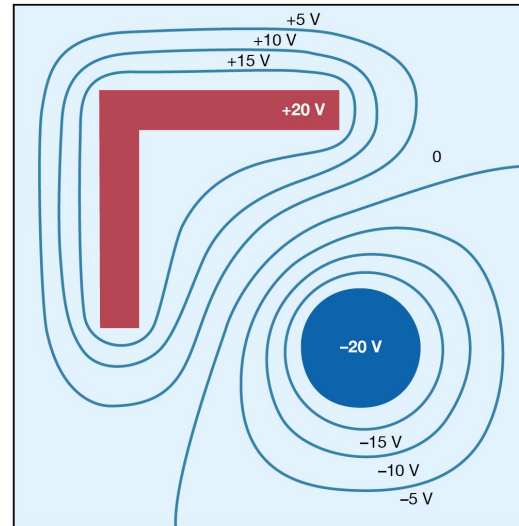
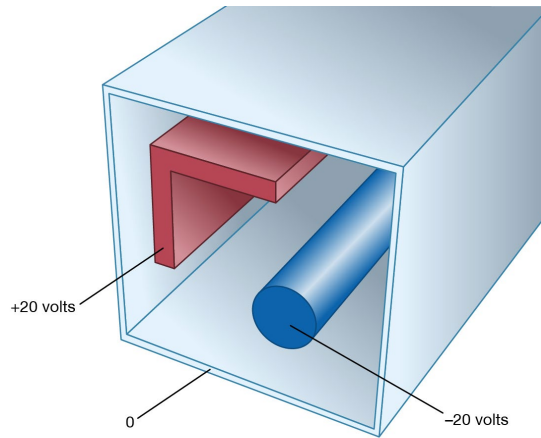
Some examples:



Two long cylindrical conductors



Large batteries for energy storage



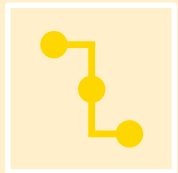
- If the potentials at the boundaries are known, geometry is not simple enough to apply the method of images; how do we solve such an electrostatic problem? For example, how to find the variation of potential V between two electrodes, let's say, in a special battery used for *large-scale storage*.
- Is there any current in electrostatic problems? How to define current at a point in space?

Learning goals

After today's lecture you will be able to



Find distribution of potential V in the space using *Method of Separation of Variables (SoV)* in the **Electrostatic Boundary Value Problems**.



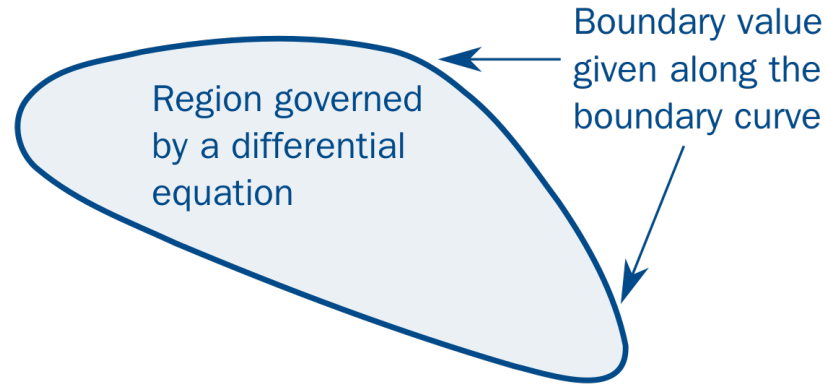
Calculate *current density J* at a point in the space using **point form of Ohm's Law**.



Calculate *resistance and capacitance* of systems **using conductivity σ and permittivity ϵ** of the material medium.

Method of Separation of Variables (SoV)

Partial differential equations (of electrostatics) with boundary conditions are called **boundary-value problems**. In such electrostatic problems, the potential on specified boundaries is known.



The 'Method of Separation of Variables' is a mathematical tool for solving the second-order partial differential equation (e.g., Laplace's equation) of boundary value problems.

Key idea: This method separates functions of multiple variables into multiple functions of single variables. For example : $V(x, y, z) = X(x)Y(y)Z(z)$

Why can we do this?

Perhaps x , y , and z at the boundaries are independent, and the solution can be separated into independent parts. If the solution found with this premise satisfies the equation and boundary conditions, then it must be the correct solution (Uniqueness theorem).

SoV for Laplace's Equation

Laplace's equation in **Cartesian coordinates**: $\vec{\nabla}^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$

When we solve it, we want to find the function $V(x, y, z)$.

In SoV, we are constructing this function, without actually solving the 2nd order PDE.

If our constructed function is correct, it will

- Satisfy Laplace's equation
- Match boundary conditions

Let's consider, $V(x, y, z) = X(x)Y(y)Z(z)$,

Substitute it in Laplace's equation:

$$Y(y)Z(z)\frac{d^2 X(x)}{dx^2} + X(x)Z(z)\frac{d^2 Y(y)}{dy^2} + X(x)Y(y)\frac{d^2 Z(z)}{dz^2} = 0$$

How does it help? We converted our PDE to ODE!

$$Y(y)Z(z)\frac{d^2X(x)}{dx^2} + X(x)Z(z)\frac{d^2Y(y)}{dy^2} + X(x)Y(y)\frac{d^2Z(z)}{dz^2} = 0 \quad (1)$$

Dividing both sides by $X(x)Y(y)Z(z)$

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} = 0 \quad (2)$$

Since x , y , and z are independent variables, each term on the LHS must be a constant,

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} = -k_x^2; \quad \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} = -k_y^2; \quad \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} = -k_z^2$$

The use of the negative sign and the square to define the constants is arbitrary.

Thus, (2) can be written as $k_x^2 + k_y^2 + k_z^2 = 0$

The constants are called *separation constants*, and they can be zero, a real number, or an imaginary number.

$$\frac{1}{X(x)} \frac{d^2 X(x)}{dx^2} = -k_x^2; \quad \frac{1}{Y(y)} \frac{d^2 Y(y)}{dy^2} = -k_y^2; \quad \frac{1}{Z(z)} \frac{d^2 Z(z)}{dz^2} = -k_z^2$$

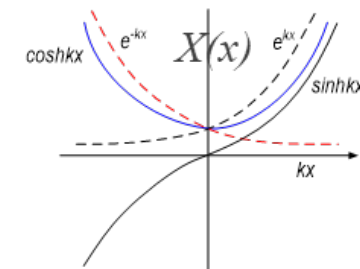
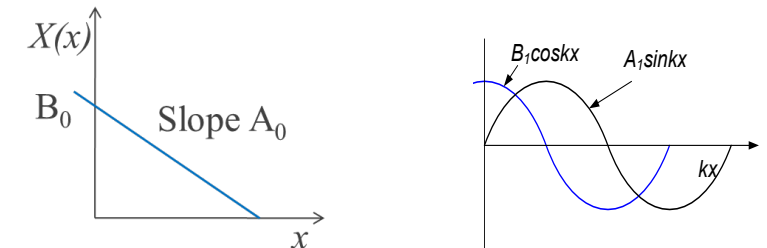


$$\frac{d^2 X(x)}{dx^2} + k_x^2 X(x) = 0; \quad \frac{d^2 Y(y)}{dy^2} + k_y^2 Y(y) = 0; \quad \frac{d^2 Z(z)}{dz^2} + k_z^2 Z(z) = 0$$

These 3 equations are completely independent and can be solved separately. They are 2nd order ODE and have standard solutions. The nature of the solution is controlled by the constant.

Standard solution of X(x), based on the possible value of k_x :

k_x^2	k_x	X(x)	
0	0	Linear	$A_0 x + B_0$
Positive	Real	Sinusoidal	$A_1 \sin k_x x + B_1 \cos k_x x$
Negative	Imaginary	Exponential	$D_1 e^{k_x x} + D_2 e^{-k_x x}$



Similarly, $Y(y)$ and $Z(z)$ can also be found based on k_y and k_z .

Once $X(x)$, $Y(y)$ and $Z(z)$ are known, the potential $V(x,y,z)$ then can be found as,

$$V(x, y, z) = X(x)Y(y)Z(z)$$

A similar process can be followed for two-dimensional cylindrical coordinate problems.

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \left(\frac{\partial^2 V}{\partial \theta^2} \right) = 0$$

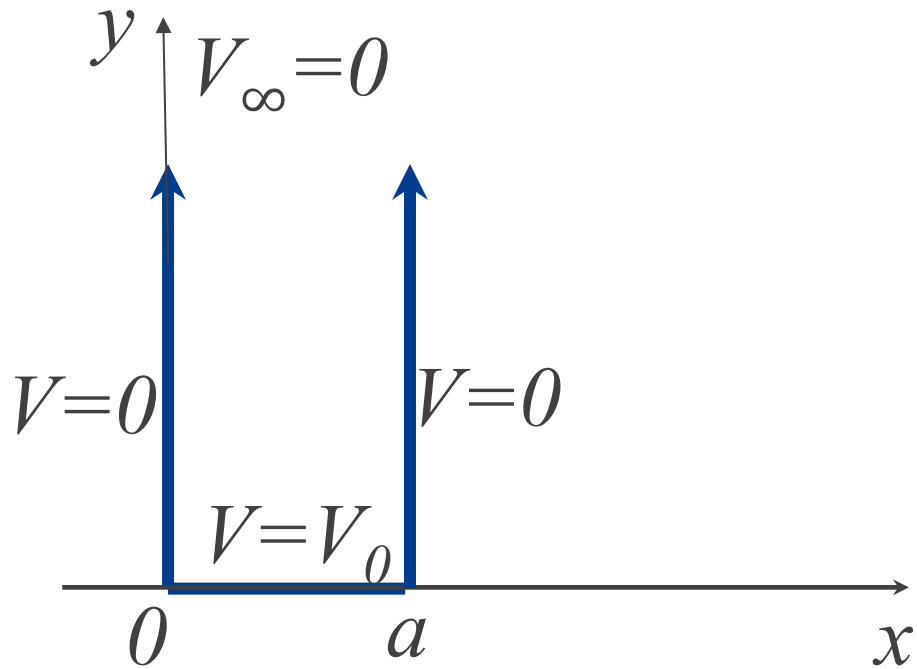
$$V(r, \theta) = R(r)\Theta(\theta)$$

$$\frac{r}{R(r)} \frac{d}{dr} \left[r \frac{dR(r)}{dr} \right] + \frac{1}{\Theta(\theta)} \frac{d^2 \Theta(\theta)}{d\theta^2} = 0$$

Solving a 3-D problem in cylindrical coordinates requires knowledge of the Bessel function.

Example 4

A two-dimensional problem is defined by two parallel surfaces, both semi-infinite in extent, and both at zero potential. The bottom surface is at potential V_0 . Also, the potential at infinity is zero. Find potential everywhere in the cavity.



Solution

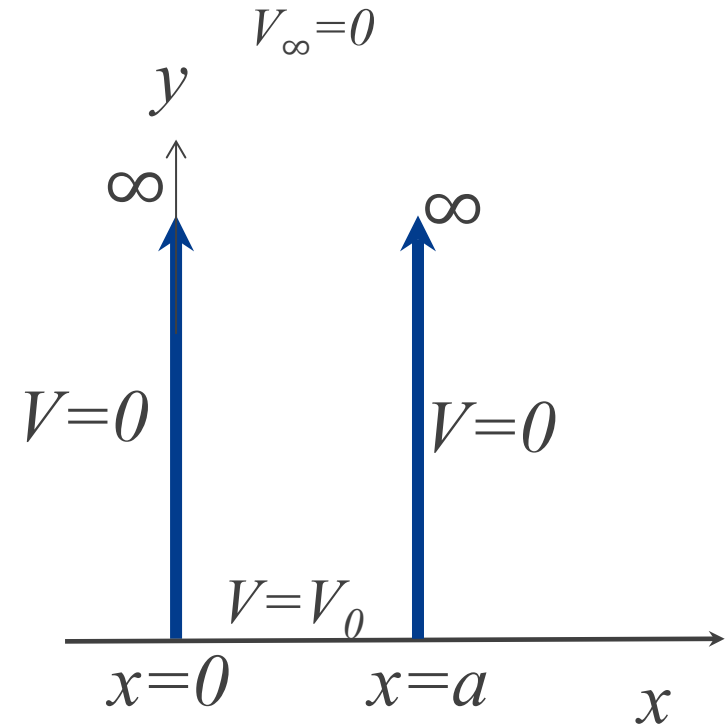
Step 1: List the boundary conditions

x -axis boundary conditions:

- (i) $V(0, y) = 0, y < \infty$
- (ii) $V(a, y) = 0, y < \infty$

y -axis boundary conditions:

- (iii) $V(x, \infty) = 0,$
- (iv) $V(x, 0) = V_0, \quad 0 < x < a$



Step 2: Define constants

It is a 2-dimensional problem in the xy plane: $\vec{\nabla}^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$ (1)

Thus, $V(x, y) = X(x)Y(y)$

Substituting in (1) and dividing both sides by $X(x)Y(y)$ $\frac{1}{X(x)} \frac{d^2 X(x)}{dx^2} + \frac{1}{Y(y)} \frac{d^2 Y(y)}{dy^2} = 0$ (2)

Separation constants: $k_x^2 + k_y^2 = 0 \rightarrow k_y^2 = -k_x^2 = -k^2$

1. k_y^2 is taken as a negative constant (k_y is imaginary), as we anticipate $Y(y)$ to be exponentially decaying because from boundary conditions at $y = 0$, $V = V_0$ and at $y = \infty$, $V = 0$.
2. k_x^2 is positive, k_x is real – $X(x)$ sinusoidal.

Thus, the general solution :

$$\begin{aligned} V(x, y) &= X(x)Y(y) = (A_1 \sin kx + B_1 \cos kx) (D_1 e^{ky} + D_2 e^{-ky}) \\ &= (A_1 \sin kx + B_1 \cos kx) (D_2 e^{-ky}) \end{aligned}$$

Constant	Function
0	Linear
Positive $\rightarrow$ Real	Sinusoidal
Negative $\rightarrow$ Imaginary	Exponential

Step 3: Find the values of constants from the boundary conditions

$$V(x, y) = (A_1 \sin kx + B_1 \cos kx) (D_2 e^{-ky})$$

At the boundaries of the x-axis,

(i) At $x = 0$, and any value of $y < \infty$

$$V(0, y) = B_1 (D_2 e^{-ky}) = 0 \rightarrow B_1 = 0$$

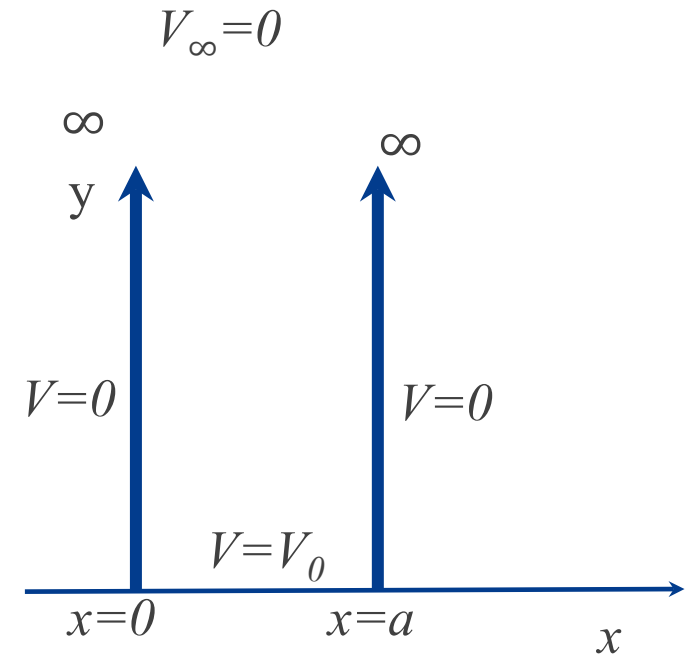
Thus, $X(x)$ is a sine function.

$$\therefore V(x, y) = (A_1 \sin kx) (D_2 e^{-ky})$$

(ii) At $x = a$, and any value of $y < \infty$

$$V(a, y) = A_1 \sin ka (D_2 e^{-ky}) = 0 \rightarrow A_1 \sin ka = 0 \rightarrow ka = n\pi \rightarrow k = \frac{n\pi}{a} \quad \text{where, } n = 1, 2, 3, \dots$$

$n = 0$ is excluded; otherwise, the separation constant k will be 0, meaning the $V(x, y)$ will be linear, but we know already that $V(x, y)$ is not linear. Negative numbers are excluded because they yield the same result with the opposite sign.



Solution at this stage is

$$V(x, y) = \left(A_1 \sin \frac{n\pi}{a} x \right) \left(D_2 e^{-\frac{n\pi}{a} y} \right) = C \left(\sin \frac{n\pi}{a} x \right) \left(e^{-\frac{n\pi}{a} y} \right)$$

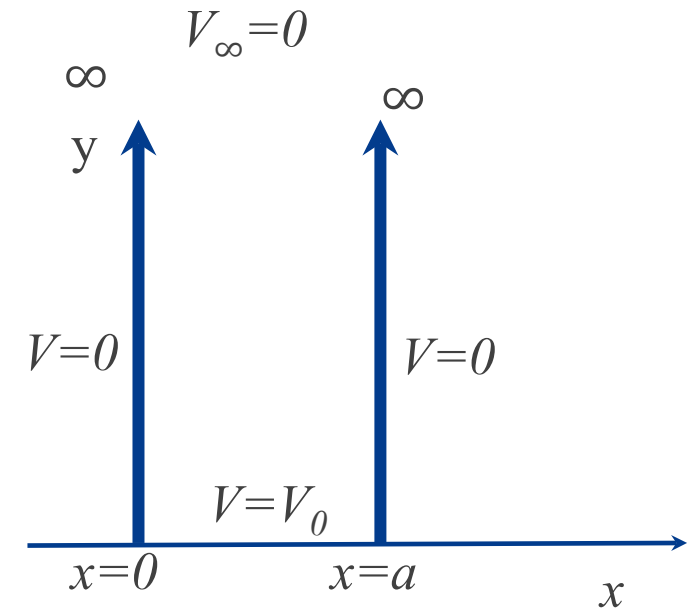
where, $C = A_1 D_2$

At the boundaries of the y-axis,

(iii) At $y = \infty$, $\rightarrow V(x, \infty) = 0 \rightarrow$ no positive exponent

On the y-axis, V reduces exponentially from V_0 to 0.

(iv) At $y = 0$, $V(x, 0) = V_0$, $0 < x < a$

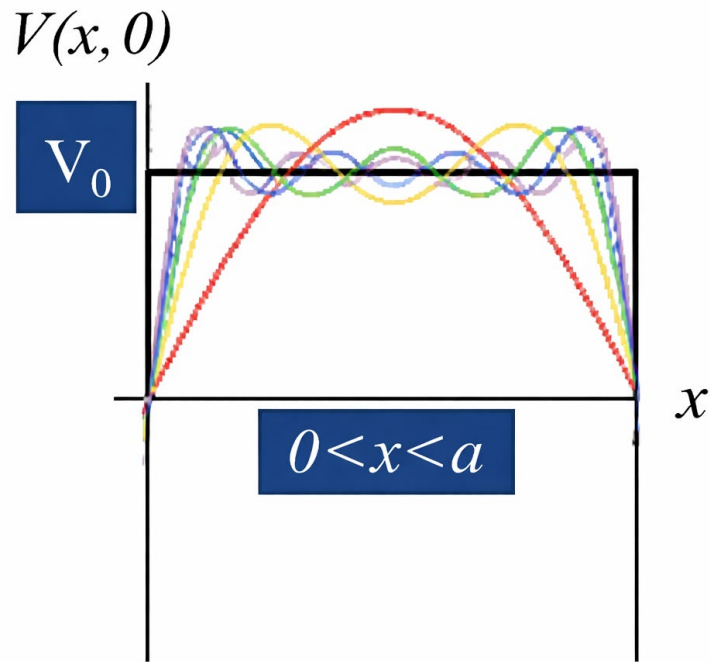


To satisfy this boundary condition, we can't simply substitute $y = 0$ in our solution. Because this will result in a sinusoidal solution, no constant C can satisfy the boundary condition (iv).

However, if we consider the superposition of solutions:

$$V(x, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{a} e^{-\frac{n\pi}{a} y}$$

Now, when $y=0$, $V(x, 0) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{a} = V_0$



$$V(x, 0) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{a} = V_0, \quad 0 < x < a$$

This is a Fourier sine series, which approximates a square waveform.

C_n : coefficients of the Fourier series, and we can find them using the procedure of finding the coefficients of a Fourier series.

Step 4: Finding of C_n

$$\sum_{n=1}^{\infty} \int_0^a C_n \sin \left(\frac{n\pi}{a} x \right) \sin \left(\frac{m\pi}{a} x \right) dx = \int_0^a V_0 \sin \left(\frac{m\pi}{a} x \right) dx$$

For the RHS,

$$\int_0^a V_0 \sin \left(\frac{m\pi}{a} x \right) dx = V_0 \int_0^a \sin \left(\frac{m\pi}{a} x \right) dx = V_0 \left[\frac{-\cos \left(\frac{m\pi}{a} x \right)}{\frac{m\pi}{a}} \right]_0^a = \frac{V_0 a}{m\pi} [-\cos m\pi + 1]$$

$$\text{Thus, } \int_0^a V_0 \sin \left(\frac{m\pi}{a} x \right) dx = \begin{cases} \frac{2aV_0}{m\pi} & \text{for } m \text{ odd} \\ 0 & \text{for } m \text{ even} \end{cases}$$

For the LHS,

$$\int_0^a C_n \sin \frac{n\pi}{a} x \sin \frac{m\pi}{a} x dx = \frac{C_n}{2} \int_0^a \left[\cos \frac{(n-m)\pi}{a} x - \cos \frac{(n+m)\pi}{a} x \right] dx$$

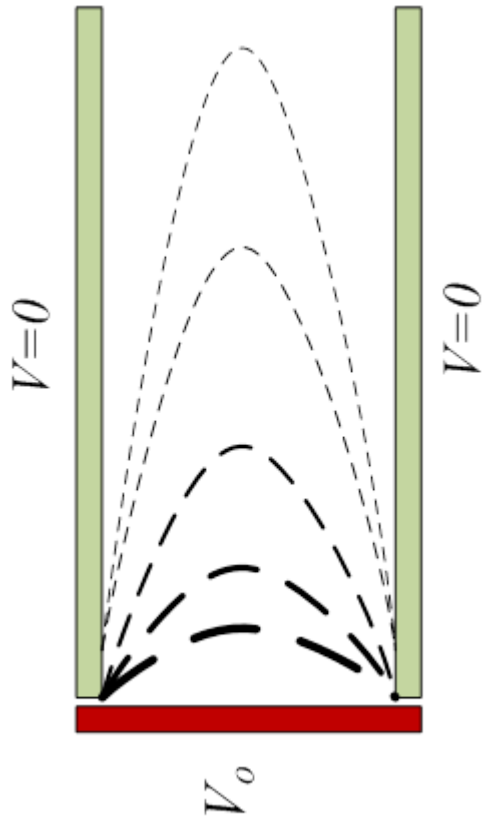
$$\text{Thus, } \int_0^a C_n \sin \frac{n\pi}{a} x \sin \frac{m\pi}{a} x dx = \begin{cases} \frac{C_n}{2} a & \text{for } m = n \\ 0 & \text{for } m \neq n \end{cases}$$

To satisfy both RHS and LHS, n must be odd and $m = n$.

$$\frac{2aV_0}{n\pi} = \frac{C_n}{2} a \rightarrow C_n = \frac{4V_0}{n\pi}$$

Step 5: Final solution and interpretation

$$V(x, y) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4V_0}{n\pi} \sin \frac{n\pi x}{a} e^{-\frac{n\pi}{a}y}$$



Potential distribution

The potential $V(x,y)$ in the cavity varies as a superposition of sine waves, the amplitudes of which decay as y approaches infinity.

The solution satisfies all the boundary conditions and Laplace's equation.

⚠ Exam Signposting — Separation of Variables

In exams, you may be asked to:

- ✓ Assume a separable solution $V(x,y) = X(x)Y(y)$
- ✓ Reduce Laplace's equation to ODEs
- ✓ Identify separation constants
- ✓ Apply boundary conditions to determine allowed values

You are expected to:

- Show the logical steps clearly
- Justify why separation constants must be positive/negative
- Construct the expected final potential function

You are NOT required to:

- ✗ Re-derive full Fourier coefficient integrals unless explicitly asked

Separation of Variables — The Roadmap

Goal

Find $V(x, y)$ that:

- Satisfies **Laplace's equation**
- Satisfies the **boundary conditions**

The 7-Step Process

- 1 **Start with Laplace's equation** $\nabla^2 V = 0$
- 2 **Assume a separable form** $V(x, y) = X(x)Y(y)$
- 3 **Substitute into Laplace's equation**
- 4 **Divide and separate variables**
→ Each side must equal a **constant**
- 5 **Solve resulting ODEs**
(Choose sine, cosine, or exponential depending on boundary type)
- 6 **Apply boundary conditions**
→ Determine allowed values of separation constants
→ Obtain discrete eigenvalues (e.g., $k = n\pi/a$)
- 7 **Superpose solutions (Fourier series)**
→ Match full boundary condition

Big Idea

Boundary conditions determine:

- The allowed functions
- The allowed constants
- The final shape of the solution

Separation of variables is not magic —

it is a structured way of building a solution that fits the boundaries.

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

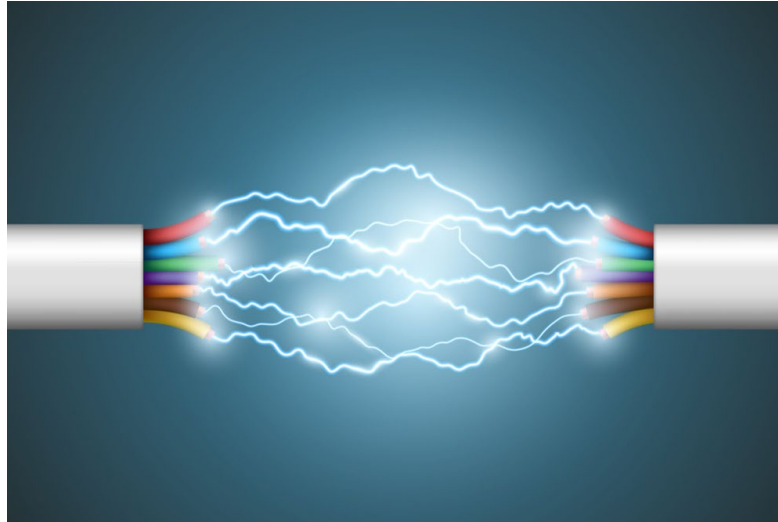
One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*

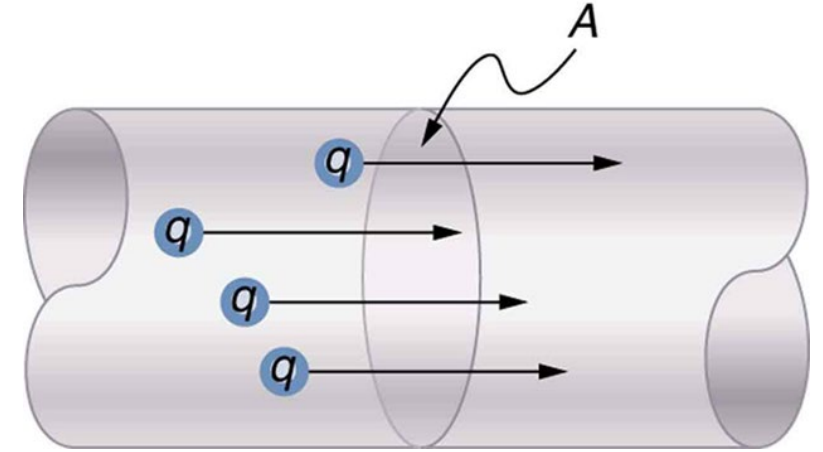


Current in electrostatic problems

Is there any current in an electrostatic field?



Current = flow of charge



$$i = \frac{dq}{dt} \rightarrow I = \frac{\Delta Q}{\Delta t}$$

When $\mathbf{E}$ is applied, the free charges inside a conducting media can move under the influence of the impressed electric field $\mathbf{E}$. - known as **Steady** or stationary current which is time-independent (DC current). Current is not a point vector: we cannot define it at a point in the space .

Current density vector $\mathbf{J}$

To analyse steady or stationary current in the space, we need to define a vector that can represent current in the space.

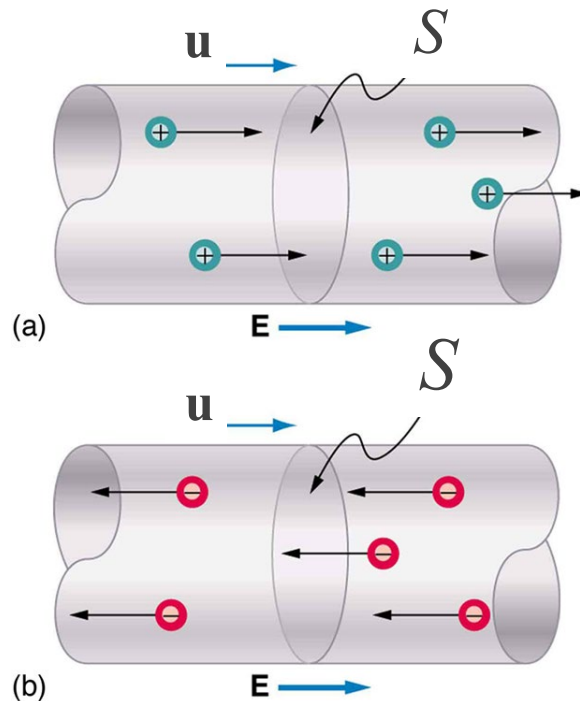
When $\mathbf{E}$ is applied, the free charges inside a conducting media move with an average velocity $\mathbf{u}$ in the parallel direction of $\mathbf{E}$.

The average distance travelled by the charges ΔQ in Δt (s) is $\Delta l = \mathbf{u} \Delta t$ (m).

If ρ is the charge density, the total charge ΔQ , which moved through an area Δs in Δt (s) in the direction of $\mathbf{E}$:

$$\Delta Q = \rho \times (\text{volume}) = \rho (\Delta \vec{l} \bullet d\vec{s}) = \rho \vec{u} \Delta t \bullet d\vec{s}$$

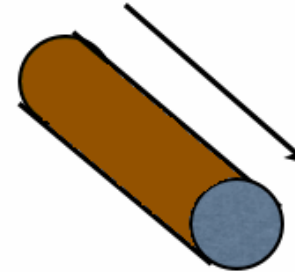
$$\Delta I = \frac{\Delta Q}{\Delta t} = \rho \vec{u} \bullet d\vec{s} \rightarrow \quad \vec{J} = \frac{\Delta I}{d\vec{s}} = \rho \vec{u} \quad \text{A/m}^2$$



Current Vs Current density

$$i = \frac{dq}{dt} \rightarrow I = \frac{\Delta Q}{\Delta t}$$

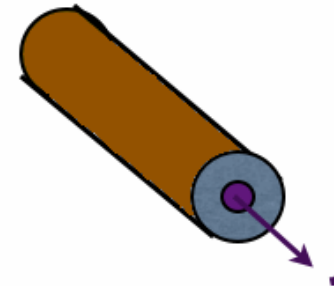
**Current I = Total Flow
of Charge Per Time [A]**



Unit is C/s or A

www.maxwells-equations.com

**Current Density J = Total Flow
of Charge Per Time [A] over a cross
section of area [m²]**



Unit is A/m²

$$\vec{J} = \frac{\Delta I}{\Delta \vec{s}} = \rho \vec{u} \quad \text{A/m}^2$$

$$\text{Total current } I = \int_s \vec{J} \bullet d\vec{s} \quad \text{A}$$

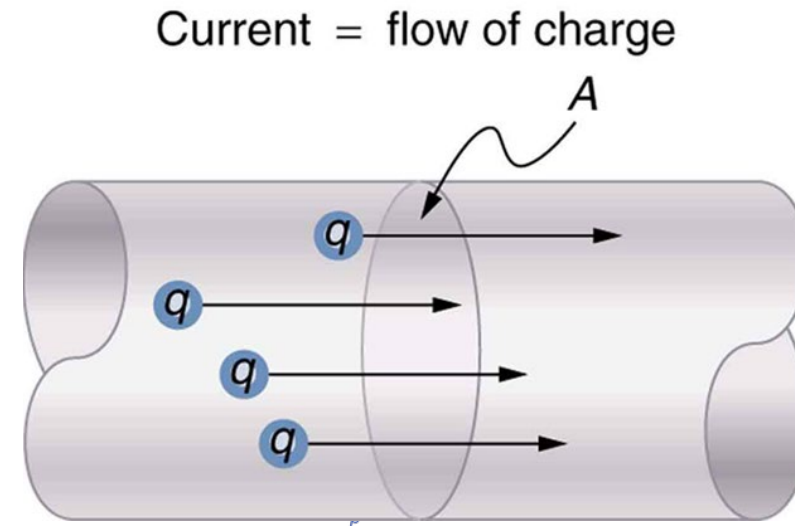
Example 5 (Application)

In a tube-shaped photovoltaic cell, cathode emits electrons when illuminated by light and an anode collects them. The electrons are emitted uniformly over the surface of the cathode, which is a disk of radius $a = 10\text{mm}$. A current of 0.01A is measured in the external circuit.

Calculate the current density and charge density next to the cathode. Assume the electrons are moving with a velocity of 100m/s .

$$J = \frac{I}{S} = \frac{I}{\pi a^2} = 31.83 \text{ A/m}^2$$

$$J = \rho u \rightarrow \rho = \frac{J}{u} = 0.3183 \text{ C/m}^3$$



Challenge questions: If we want to maintain the same current and charge density near the cathode but the external circuit requires 0.05A , what changes will you make to the cell?

Conductivity and Ohm's law in vector form

For a conducting material, $\vec{u} \propto \vec{E} \rightarrow \vec{u} = \mu_e \vec{E}$

The proportionality constant μ_e is the *mobility of charge carrier* in $\text{m}^2/\text{V}\cdot\text{s}$.

$$\vec{J} = \rho \vec{u} = \rho \mu_e \vec{E} = \sigma \vec{E} \quad \text{A/m}^2$$

$\sigma = \rho \mu_e$ is called conductivity of the medium; its unit is $\text{A/V}\cdot\text{m}$ or Siemens/m (S/m) or $(\Omega\cdot\text{m})^{-1}$.

Reciprocal of conductivity $\frac{1}{\sigma}$ is the resistivity in $\Omega\cdot\text{m}$.

$$\vec{J} = \sigma \vec{E}$$

This is the point from of **Ohm's law** ($V = IR$ or $I = V/R = G V$).

Resistance of a conductor with uniform cross-section

Potential difference,

$$V_{12} = -\int_l \vec{E} \cdot d\vec{l} = El \rightarrow E = \frac{V_{12}}{l}$$

Total Current,

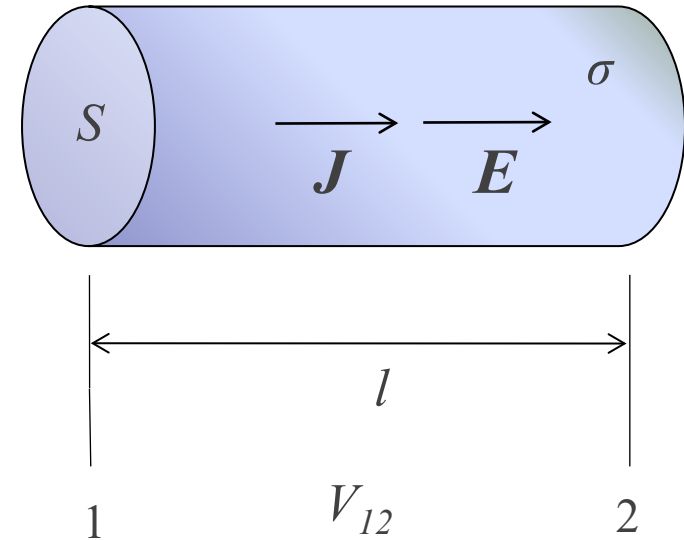
$$I = \int_s \vec{J} \cdot d\vec{s} = JS \rightarrow J = \frac{I}{S}$$

Using point form Ohm's law,

$$\vec{J} = \sigma \vec{E} \quad A/m^2; \quad \frac{I}{S} = \sigma \frac{V_{12}}{l} \rightarrow V_{12} = \left(\frac{l}{\sigma S} \right) I$$

Comparing with $V_{12} = IR$

$$\therefore R = \frac{l}{\sigma S} \quad (\Omega) \text{ or } R = \text{resistivity} \times \frac{l}{S}$$



Practice Example 6

A DC fuse made of silver alloy is shown in the figure. The safe current density J of the alloy is 10^6 A/m².

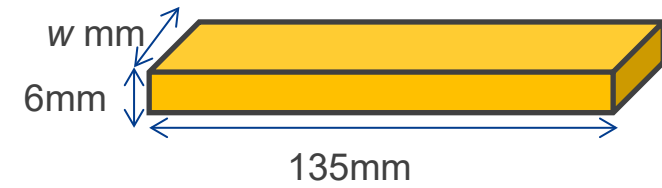
- (i) Design the width w of the fuse link so that it will break if the current is higher than 160A.

Ans: 26.7mm

- (i) Calculate the resistance of the fuse if the conductivity of the silver alloy is $\sigma = 6.1 \times 10^7$ S/m.

Ans: 14.2 $\mu\Omega$

Dimensions of the link



Power and power density

$$P = VI = \left(\int_L \vec{E} \bullet d\vec{l} \right) \left(\int_s \vec{J} \bullet d\vec{s} \right) = \int_{vol} \vec{E} \bullet \vec{J} dv \text{ [W]} = (\vec{E} \bullet \vec{J}) \int_{vol} dv$$

The power density (Watts per unit volume):

$$p_d = \vec{E} \bullet \vec{J} = \frac{|\vec{J}|^2}{\sigma} \text{ [W/m}^3\text{]}, \quad \because \vec{J} = \sigma \vec{E}$$

⚠ Exam Signposting — Current Density/Ohm's Law

You may be required to:

- ✓ Compute J from E
- ✓ Integrate J to find the total current
- ✓ Distinguish between I and J
- ✓ Apply $J = \sigma E$ correctly

Common mistakes:

- ✗ Treating J as a current instead of a vector
- ✗ Forgetting units (A/m² vs A)

Marks are awarded for:

- Correct vector interpretation
- Clear unit handling

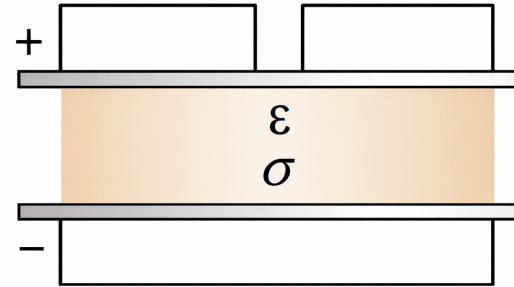
Can you find the power density of the fuse in the practice example?

Resistance and capacitance of homogeneous medium

(Poor conductor or lossy dielectric)

We know for capacitance,

$$C = \frac{Q}{V} = \frac{\oint_S \vec{D} \cdot d\vec{s}}{-\int_L \vec{E} \cdot d\vec{l}} = \frac{\oint_S \epsilon \vec{E} \cdot d\vec{s}}{-\int_L \vec{E} \cdot d\vec{l}}$$



⚠ Exam Signposting — RC Relation

For homogeneous media:

$$RC = \frac{\epsilon}{\sigma}$$

You may be asked to:

- ✓ Relate material properties to time constants
- ✓ Predict behaviour of lossy dielectric
- ✓ Compare resistance and capacitance structurally

You are expected to understand:

Physical meaning of ϵ/σ

How conductivity affects decay

and for resistance

$$R = \frac{V}{I} = \frac{-\int_L \vec{E} \cdot d\vec{l}}{\int_S \vec{J} \cdot d\vec{s}} = \frac{-\int_L \vec{E} \cdot d\vec{l}}{\int_S \sigma \vec{E} \cdot d\vec{s}}$$

$$\therefore RC = \frac{\epsilon}{\sigma}$$

Thus, if C is known, R can be found and vice versa provided σ and ϵ of the material are known.

Equation of continuity

In electrostatics, if charges are moving from a charged body, charges reduces. For example, during a lightning strike, charges decay from the cloud.

$$I = \oint_s \vec{J} \cdot d\vec{s} \text{ and } Q = \int_{vol} \rho dv$$

$$I = -\frac{dQ}{dt},$$

$$\therefore \oint_s \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \int_{vol} \rho dv \rightarrow \int_{vol} \vec{\nabla} \cdot \vec{J} dv = -\int_{vol} \frac{\partial \rho}{\partial t} dv$$

Divergence theorem

$$\therefore \vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

Equation of continuity



$$\vec{\nabla} \bullet \vec{J} = -\frac{\partial \rho}{\partial t}, \quad \vec{J} = \sigma \vec{E}$$

Substituting $\vec{J}$, $\sigma \vec{\nabla} \bullet \vec{E} = -\frac{\partial \rho}{\partial t}$, $\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon}$

Substituting $\vec{\nabla} \bullet \vec{E}$, and rearranging, $\therefore \sigma \frac{\rho}{\epsilon} + \frac{\partial \rho}{\partial t} = 0 \rightarrow \rho = \rho_0 e^{-\left(\frac{\sigma}{\epsilon}\right)t}$



This is a first-order differential equation, and a standard solution exists in terms of initial charge ρ_0

If charge density ρ does not vary with time (no decaying),

$$\vec{\nabla} \bullet \vec{J} = 0 \rightarrow \oint_s \vec{J} \cdot d\vec{s} = 0 \rightarrow \sum_n I_n = 0 \quad \text{Kirchhoff's current law (KCL)}$$

$$\vec{\nabla} \times \vec{E} = 0 \Rightarrow \vec{\nabla} \times \frac{\vec{J}}{\sigma} = 0 \rightarrow \vec{\nabla} \times \vec{J} = 0$$

⚠ Exam Signposting — Continuity Equation

You may be asked to:

- ✓ Interpret $\nabla \cdot \vec{J}$ physically
- ✓ Relate time-varying charge to current flow
- ✓ Derive special case when charge is steady

If ρ is constant in time:

$$\nabla \cdot \vec{J} = 0$$

This connects directly to Kirchhoff's Current Law.

Marks are awarded for:

Physical interpretation

Clear reasoning

Homework (practice example 7)

A lightning strike lasts for $250 \mu\text{s}$ and triggers a current $I = 25\text{kA}$ in a cylindrical channel of radius $r = 3\text{mm}$ and length $l = 2\text{km}$. Assume the conductivity in the channel is $\sigma = 2.2 \times 10^5 \text{S/m}$ and all energy is dissipated within the channel.

(a) Calculate the total power dissipated in the lightning strike.

A: 200 GW

(a) Calculate the power density in the channel in W/m^3 .

A: $3.54 \times 10^{12} \text{W/m}^3$

(c) Calculate electric field intensity E in the channel. **A: 4.02kV/m**

(d) What is the total energy dissipated? (Energy is power integrated over time). **A: 50 MJ**

You may find following formula useful:

$$R = \frac{l}{\sigma S}, \quad p_d = \frac{\text{Power}}{\text{Volume}} = \vec{E} \bullet \vec{J}, \quad \vec{J} = \sigma \vec{E}$$

Boundary conditions:

$$E_{1t} = E_{2t}$$



$$\frac{J_{1t}}{\sigma_1} = \frac{J_{2t}}{\sigma_2}$$

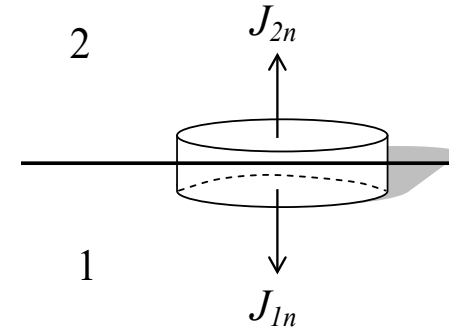
$$J_{1t} = \sigma_1 E_{1t}, \quad J_{2t} = \sigma_2 E_{2t}$$

$$\vec{D}_{1n} - \vec{D}_{2n} = \epsilon_1 \vec{E}_{1n} - \epsilon_2 \vec{E}_{2n} = \rho_s$$



$$J_{1n} = \sigma_1 E_{1n}, \quad J_{2n} = \sigma_2 E_{2n}$$

$$\epsilon_1 \frac{\vec{J}_{1n}}{\sigma_1} - \epsilon_2 \frac{\vec{J}_{2n}}{\sigma_2} = \rho_s \quad (1)$$

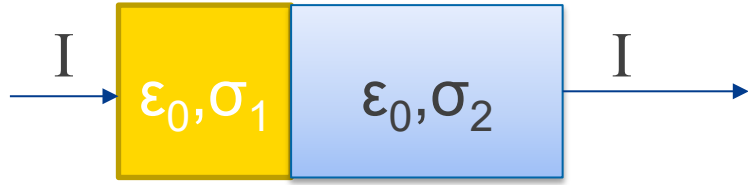


$$\vec{\nabla} \bullet \vec{J} = 0 \rightarrow J_{1n} = J_{2n} \quad (2) \text{ (KCL)}$$

From (1) and (2), we can conclude that if ρ_s is zero,

$$\sigma_2 / \sigma_1 = \epsilon_2 / \epsilon_1$$

Application Example 8 (Boundary conditions)



Two poorly conducting materials of surface area S are connected to a current source of I . Calculate the surface charge density between the two conductors.

$$J_1 = J_2 = \frac{I}{S}$$

$$E_1 = \frac{J_1}{\sigma_1} = \frac{I}{S\sigma_1}, \quad E_2 = \frac{J_2}{\sigma_2} = \frac{I}{S\sigma_2}$$

$$D_1 - D_2 = \rho_s \text{ (from interface condition of } D_n \text{)}$$

$$\therefore \rho_s = \epsilon_0 (E_1 - E_2) = \frac{\epsilon_0 I}{S} \left(\frac{1}{\sigma_1} - \frac{1}{\sigma_2} \right)$$

Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

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1 of 3

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Scan the QR or use link to join



<https://forms.office.com/r/vJs4eS-WdQH>

 Copy link

What You Must Be Able To Do After Today

1. Use separation of variables to reduce Laplace's equation.
 2. Apply boundary conditions to determine allowed separation constants.
 3. Recognize when superposition (Fourier series) is required.
 4. Apply $J = \sigma E$ correctly.
 5. Interpret the continuity equation physically.
 6. Relate resistance and capacitance through material properties.
- If you can do these six things, you are exam-ready for this topic.

Review and further practice (Recommended)

Examples 4-7, 4-8, and 4-9 are from the textbook of D.K. Cheng.

Examples 5-2, 5-4, 5-5, and 5-6 from the textbook of D.K. Cheng.